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Ecole Nationale des Sciences Appliquées - Safi

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National School of Applied Sciences of Safi
Department of Computer Science, Networks, and Telecommunications

Internship Report:

Telco cloud design, IaaS and service connectivity.

Internship conducted at:

orange™ Business

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I would like to express my deepest gratitude to Orange Business Services for offering me the opportunity to complete my internship within their organization.

My sincere thanks go to my internship supervisor Ayyoub Demami and the entire technical team for their guidance, patience, and constant support throughout this experience. Their expertise and advice helped me better understand how theoretical concepts are applied in real-world network infrastructures.

I would also like to thank all the engineers and staff members who contributed to making this internship a valuable and enriching experience.

Finally, I would like to extend my appreciation to my professors and the administrative staff at ENSA Safi, whose teachings and encouragement gave me the knowledge and confidence to successfully complete this internship.

General Introduction

This internship was carried out within Orange Business services as part of the engineering curriculum at ENSA Safi.
The main objective was to gain practical experience in advanced network configuration and to strengthen the theoretical knowledge acquired during the academic program.

Throughout this internship, I designed and implemented a multi-protocol network topology using EVE-NG on VMware, reproducing a service provider style backbone. The project integrated several core technologies such as IP, OSPF, MPLS, and BGP, while also exploring route reflection and BGP path control mechanisms.

In addition to the lab work, I had the opportunity to explore the telco cloud, IaaS (Infrastructure as a Service), and service connectivity, which correspond to the specialization area of my internship supervisor. This exposure helped me understand how cloud infrastructure and traditional networking intersect within modern telecom environments.

This experience allowed me to deepen my understanding of how modern service provider networks operate, enhance my technical and troubleshooting skills, and apply professional practices in a real-world context.

Introduction General

Ce stage a été réalisé au sein d'Orange business services, dans le cadre du cursus d'ingénierie à l'ENSA Safi.
L'objectif principal était d'acquérir une expérience pratique en configuration avancée de réseaux et de renforcer les connaissances théoriques acquises durant la formation académique.

Au cours de ce stage, j'ai conçu et mis en œuvre une topologie réseau multi protocoles à l'aide de EVE-NG sur VMware, reproduisant une infrastructure de type opérateur. Le projet a intégré plusieurs technologies essentielles telles que IP, OSPF, MPLS et BGP, tout en explorant les mécanismes de route r5eflection et de contrôle des chemins BGP.

En plus du travail en laboratoire, j'ai eu l'occasion d'explorer le Telco Cloud, l'Infrastructure as a Service (IaaS) et la connectivité des services, qui correspondent au domaine de spécialisation de mon encadrant de stage. Cette expérience m'a permis de comprendre comment les infrastructures cloud et les réseaux traditionnels s'articulent dans les environnements télécoms modernes.

Cette expérience m'a permis d'approfondir ma compréhension du fonctionnement des réseaux d'opérateurs modernes, de perfectionner mes compétences techniques et de dépannage, et d'appliquer des pratiques professionnelles dans un contexte réel.

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List of abbreviations

Acronyms	Signification
AS	Autonomous System
BGP	Border Gateway Protocol
CE	Customer Edge (Router)
CPU	Central Processing Unit
DHCP	Dynamic Host Configuration Protocol
DNS	Domain Name System
eBGP	External Border Gateway Protocol
FEC	Forwarding Equivalence Class
GE	Gigabit Ethernet
IaaS	Infrastructure as a Service
iBGP	Internal Border Gateway Protocol
ICMP	Internet Control Message Protocol
IGP	Interior Gateway Protocol
IP	Internet Protocol
ISP	Internet Service Provider
LDP	Label Distribution Protocol
LFIB	Label Forwarding Information Base
LSP	Label Switched Path
MPLS	Multiprotocol Label Switching
MP-BGP	Multiprotocol Border Gateway Protocol
NAT	Network Address Translation
NFV	Network Function Virtualization

OSPF	Open Shortest Path First
PE	Provider Edge (Router)
QoS	Quality of Service
RAM	Random Access Memory
RR	Route Reflector
SDN	Software-Defined Networking
SP	Service Provider
TE	Traffic Engineering
VRF	Virtual Routing and Forwarding
WAN	Wide Area Network

Chapter I: Introduction

I .1 Context of the Internship

As part of the Network and Telecommunication Engineering program at the National School of Applied Sciences of Safi (ENSA Safi), this internship represents an essential step in bridging academic knowledge with real-world industry practice.

The internship was carried out within the Cloud and Service Connectivity Department of Orange business services, a team responsible for designing and maintaining virtualized telecommunication infrastructures. This environment provided exposure to cutting-edge technologies, including Telco cloud, IaaS, and service connectivity.

The experience took place in a context where network and telecommunication infrastructures are increasingly converging with cloud computing paradigms through the adoption of Network Function Virtualization (NFV) Software Defined Networking (SDN).

This immersion allowed me to understand how operators leverage virtualization and automation to optimize network performance, service delivery, and scalability.

I.2 Objectives of the Internship

The main objective of this internship was to gain a deeper understanding of how cloud and networking technologies are integrated within telecommunication environments. Specifically, the goals included:

- ✓ Understanding the principles of cloud-based network architecture.
- ✓ Exploring the deployment and management of virtualized network services.
- ✓ Learning about the mechanisms ensuring connectivity, scalability, and service quality in Telco Cloud environments.
- ✓ Strengthening professional and technical skills through hands-on experimentation and observation.

I.3 Motivation and Personal Goals

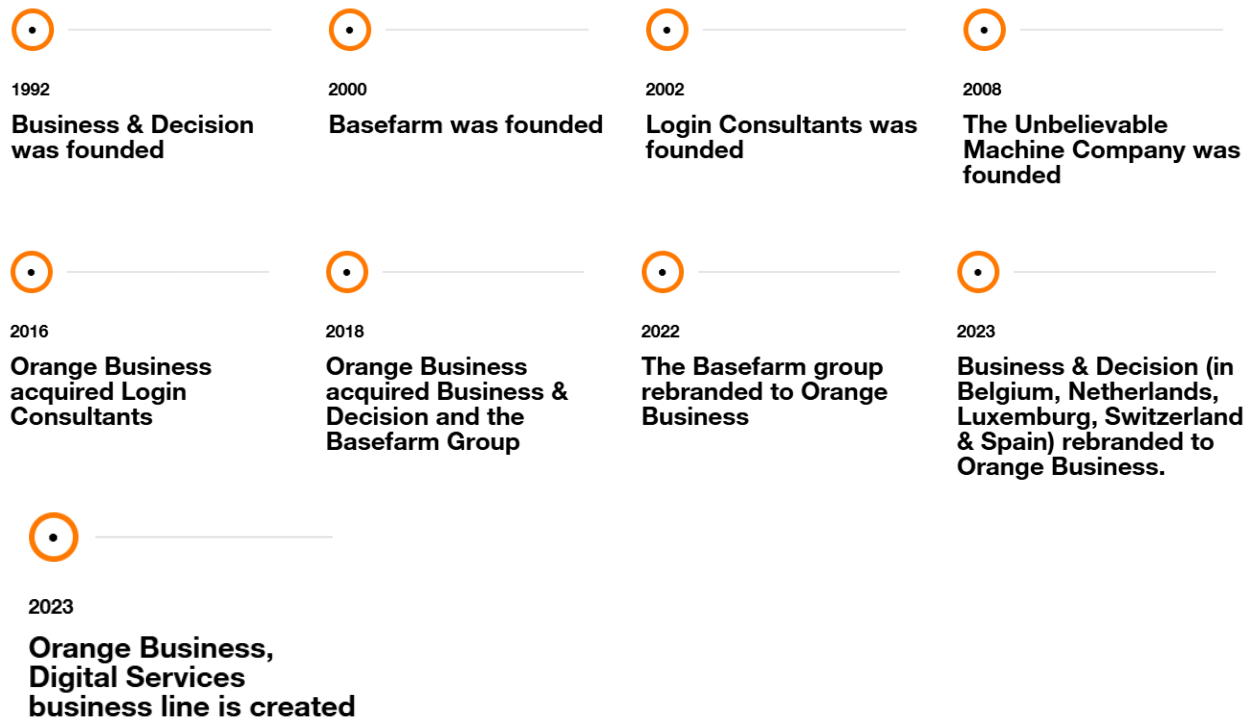
This internship was a valuable opportunity to bridge the gap between academic learning and industrial practice. Motivated by a strong interest in networking, virtualization, and cloud technologies, I aimed to explore how telecom operators leverage these technologies to provide reliable and scalable connectivity services.

It also allowed me to develop a clearer vision of my future career direction, particularly toward cloud engineering, DevOps, or network automation.

Chapter II: Company Overview

II.1 Presentation of Orange Business Services

Orange Business Services (now branded Orange Business) is the enterprise-oriented arm of the Orange Group, positioned as a leading global integrator of IT and communications solutions for multinational corporations. It was established on 1 June 2006 via the consolidation of France Télécom's existing business divisions (the Equant and Wanadoo businesses) into a single global entity. Since its founding, Orange Business has emphasized supporting enterprises in their digital transformation, leveraging the Group's telco heritage and IT services expertise to create "sustainable digital transformation" for global organizations.



Today Orange Business operates on an extremely broad international footprint. The Orange network spans more than 200 countries and territories worldwide, and the company maintains operations or alliances in roughly 65 countries to provide local teams for deployment and support. The organization employs on the order of 30,000 people globally, of whom approximately 9,000+ are designated "digital experts" specializing in areas such as networking, cloud, data, IoT and cybersecurity. This includes engineers, consultants, and researchers who design and manage large-scale enterprise networks and cloud systems. With this extensive workforce and global coverage, Orange Business supports more than 30,000 business-to-

business customers worldwide, focusing on improving efficiency, productivity and security in international operations.

Orange Business maintains a deep infrastructure to support its offerings. It operates over 70 cloud data centers across five continents and maintains 6 major support and operations centers worldwide. The company also advertises having the “No. 1 voice and data network worldwide,” reflecting its extensive MPLS/IP backbone and peering agreements in hundreds of countries. This underpins its core networking services: for example, the firm offers managed IP/MPLS WANs, flexible SD-WAN solutions, Internet and VPN connectivity, and global IoT connectivity (via its managed cellular IoT platform). In one recent initiative, Orange Business provided global IoT mobile connectivity for Lynk & Co’s “always-connected” car across Europe, demonstrating the company’s ability to deliver carrier-grade network services on a multinational scale.

In addition to networking, Orange Business offers a broad portfolio of IT services and solutions. Its cloud services include public IaaS/PaaS offerings and private/hybrid cloud deployments, along with on-demand VPN and SD-WAN solutions to integrate cloud and network. The company’s unified communications solutions cover enterprise VoIP, telepresence and collaboration platforms. Cybersecurity is handled through its subsidiary Orange Cyberdefense; the firm provides a full suite of security services (SOC, threat detection and response, managed DDoS protection, etc.) under the Orange umbrella. In short, Orange Business integrates networking, cloud, security and application platforms to offer end-to-end service connectivity and digital infrastructure. This one-stop model is explicitly aimed at accelerating the digitalization of enterprise customers, a key strategic focus for the company.

Orange Business works closely with major technology partners to enhance its offerings. It maintains strategic partnerships with AWS, Cisco, Microsoft (among others) to co-develop solutions and integrate third-party technologies. These alliances allow Orange Business to combine its global network and systems integration skills with best-of-breed cloud and networking platforms. The company serves clients across a wide range of industries: for example, analyst reports note Orange Business’ involvement in manufacturing, finance, healthcare and the public sector, and it has flagship engagements in the automotive and mobility sectors (such as the connected-car deployment for Lynk & Co).

In summary, Orange Business positions itself as a trusted partner for large enterprises in their digital transformation journey, leveraging its global reach, large expert workforce, extensive infrastructure and strong ecosystem of partners.

Figure 1: Orange business logo

II.2 Organizational Structure

Orange Business Services operates as a large international organization, structured into several departments to ensure efficient management of its diverse services. Orange Business departments specialize in helping companies with digital transformation through expertise in areas like cloud computing, cybersecurity, and data, AI, and IoT. Other specialties include creating integrated communication and collaboration solutions, building secure and automated networks (like SD-WAN and 5G), and providing customer experience solutions and digital transformation consulting. The company also offers operational support and IT services to manage and integrate complexity for enterprise clients.
The main departments include:

- Cloud: Designing, building, and managing secure, scalable, and automated cloud infrastructure across multi-cloud environments (AWS, Azure, GCP).
- Data, AI, and IoT: Helping companies become data-driven with services for data management, intelligence, and AI modeling. They have a specific platform, Datavenue, for IoT, AI, and analytics, as well as expertise in specialized areas like smart cities.
- Cybersecurity: Securing the enterprise and building a trusted digital environment through end-to-end security management and consulting.
- Networks: Providing integrated communication products and services, including secure and virtualized networks like SD-WAN and SASE, and managing global network infrastructure.
- Consulting and Digital Transformation: Guiding businesses through their digital transformation with consulting services that simplify operations and focus on growth.
- Customer and Employee Experience: Creating enhanced, omnichannel customer experience and improving the employee experience through collaboration tools and support.
- Operational and IT Services: Offering end-to-end enterprise integration and management for data centers, networks, and IT equipment, including 24/7 global support.

- Unified Communications and Collaboration: Delivering and managing solutions like IP voice services, PBX, and collaboration platforms.

II.3 Department of Assignment

During my internship, I was assigned to the Telco cloud, IaaS and service connectivity department, which focuses on the design and optimization of network architectures and service connectivity for enterprise clients.
My supervisor's domain of expertise was Telco Cloud design, Infrastructure as a Service (IaaS), and service connectivity. Under his guidance, I had the opportunity to explore the integration between traditional IP/MPLS networking and cloud-based infrastructures.

This environment allowed me to apply theoretical networking knowledge in a professional context while gaining insight into how service providers combine physical and virtual network components to deliver flexible and scalable solutions.

Chapter III: Telco Cloud, IaaS, and Service Connectivity

III.1 Introduction to Telco Cloud

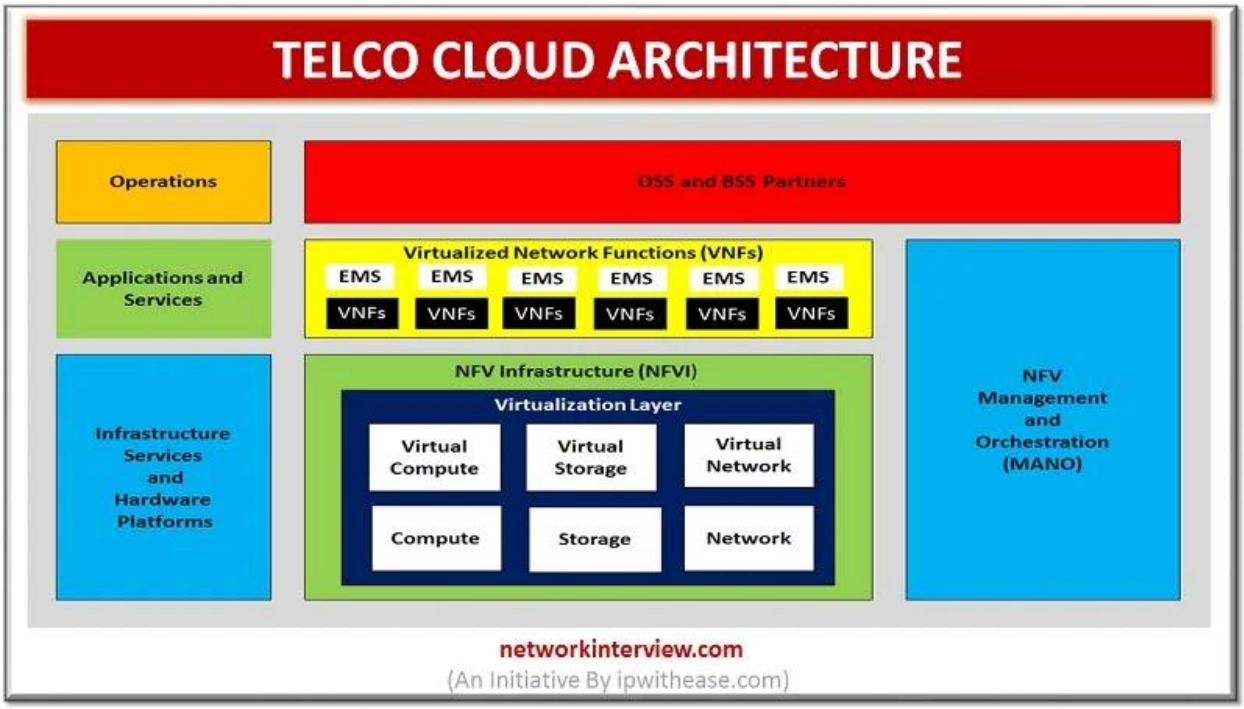


Figure 2:Telco cloud architecture

1.1 Definition of Telco Cloud

Telco cloud represents the Data Center resources which are required to deploy and manage a mobile phone network. Telco clouds are based in private Data Center facilities which are used to manage the telecommunication requirements of 3G/4G and LTE networks. With the current roll-out of 5G equipment across the mobile service provider, vendors have adopted strategies related to network function virtualization (NFV) and software-defined data center (SDDC) management which have become indispensable part of a telco setup.

1.2 Definition of Network Function Virtualization (NFV)

Network functions virtualization (NFV) is a way to virtualize network services, such as routers, firewalls, and load balancers, that have traditionally been run on proprietary hardware or dedicated hardware. NFV's functions like routing, load balancing and firewalls are packaged as virtual machines (VMs). NFV doesn't depend on dedicated hardware for each network function. NFV improves scalability and agility by allowing service providers to deliver new network services and applications on demand, without requiring additional hardware resource.

1.3 NFV Architecture – A Telco Cloud Architecture

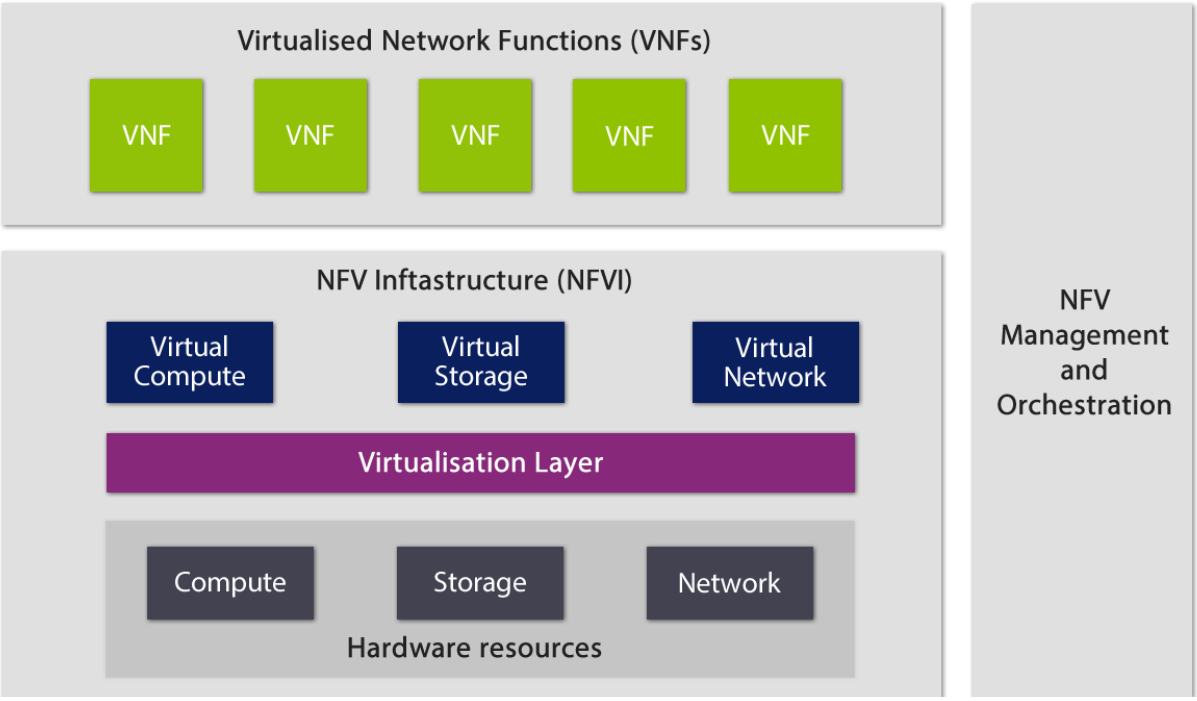


Figure 3:High-level NFV framework

The NFV architecture was proposed by the European Telecommunications Standards Institute (ETSI) which has helped to define standards for NFV implementation. Each component of the architecture is based on these standards with approach to promote better stability and interoperability. NFV architecture consists of:

1. Virtualization Network Function (VNF) Layer
2. NFV Infrastructure (NFVI) Layer
3. Operation Support Subsystem (OSS) Layer
4. Management, Automation and Network Orchestration (MANO) Layer

❖ **Virtualization Network Function (VNF) Layer**

Virtualized network functions (VNFs) are software applications that deliver network functions such as file sharing, directory services, and IP configuration. Virtual Network Function (VNF) is the key component of NFV Architecture. It virtualizes network function e.g. when a router is virtualized, it is known as a Router VNF and when a base station is virtualized it is known as a base station VNF. Similarly, it can be DHCP server VNF and Firewall VNF. When one sub-function of a network element is virtualized, it is known as a VNF. VNF is deployed on

Virtual Machines (VMs). A VNF can be deployed on multiple VMs where each VM host performs a single function of VNF. However, the whole VNF can also be deployed on a single VM as well.

Element Management System (EMS) has capability for the functional management of VNF. It includes:

- Fault management
- Configuration management
- Accounting management
- Performance and Security Management.

Depending on infrastructure, there can be provision of one EMS per VNF or else one EMS that can manage multiple VNFs. EMS can be deployed as Virtual Network Function (VNF).

❖ NFV Infrastructure (NFVI) Layer

Network functions virtualization infrastructure (NFVi) consists of the infrastructure components (compute, storage and networking) on a platform to support software, such as a hypervisor (like KVM) or a container management platform, needed to run network apps. NFV Infrastructure refers to the hardware and software components which build up the environment where VNFs are deployed, managed and executed. NFV Infrastructure includes following:

- Hardware Resources
- Virtualization Layer
- Virtual Resources

❖ Operation Support Subsystem (OSS)/Business Support System (BSS) Layer

OSS deals with network management, fault management, configuration management and service management.

❖ Management, Automation and Network Orchestration (MANO) Layer

Management, Automation and Network Orchestration (MANO) is responsible for managing NFV infrastructure and provisioning new VNFs. Management and Orchestration Layer is also abbreviated as MANO. Three components of this layer:

- Virtualized Infrastructure Manager

- VNF Manager
- Orchestrator

MANO interacts with both NFVI and VNF layer. It manages all the resources in the infrastructure layer and creates and deletes resources and manages their allocation of the VNFs.

- **Benefits of using NFV**
  - Reduced physical space required for network hardware.
  - Reduced network power consumption.
  - Reduced network maintenance costs.
  - Easier network upgrades.
  - Longer life cycle for network hardware.
  - Reduced maintenance and hardware costs.
  - Reduced physical hardware requirement.
  - Increased flexibility to run VNFs across different servers or move them around as needed when demand changes.
  - If the function is no longer needed, the VM can be decommissioned.
- **Application of NFV**
  - Mobile Edge Computing, this technology was born from the ongoing rollouts of 5G networks. The MEC uses individual components in its architecture which are similar to the NFV.
  - Software Defined Wide Area Network (SD-WAN).
  - Virtual Customer Premise Equipment (vCPE).
  - Pre-NFV/SDN-based virtualized legacy infrastructure equipment.
  - NFV telco Data Centers using SDN controllers.
  - Evolved Packet Core (EPC).
- **Conclusion**

Telco Cloud refers to a Private Cloud within a Telco/ISP environment. VNF functions are an imperative part of Telco Cloud and not extensively employed in enterprise Cloud or IT Cloud.

III.2 Introduction to IaaS

❖ **Overview**

Infrastructure-as-a-service (IaaS), also known as cloud infrastructure services, is a form of cloud computing in which IT infrastructure is provided to end users through the internet. IaaS is commonly associated with serverless computing.

❖ **IaaS vs. PaaS vs. SaaS**

IaaS is 1 of 3 widely recognized cloud service models alongside Platform-as-a-Service (PaaS) and Software-as-a-Service (SaaS) that gives users all the benefits of on-premise computing resources without the overhead. In the IaaS model, users handle the applications, data, operating system, middleware, and runtimes.

The IaaS vendor provides the virtualization, storage, network, and servers. This way, the user doesn't have to have an on-premise datacenter and doesn't have to worry about physically updating or maintaining these components themselves.

In most cases, the IaaS user has complete control of the infrastructure through an application programming interface (API) or dashboard. As the most flexible as-a-Service cloud model, IaaS makes it easier to scale, upgrade, and add resources like cloud storage instead of having to anticipate future needs and pay costs up front.

The term *as-a-Service* generally means a cloud computing service that is managed for you so that you can focus on what's more important, like your code and relationships with your customers. These offerings have grown in popularity because of their pricing models, which can be more cost-effective than self-managed, on-premise infrastructure. At the same time, the ecosystem of available technologies, partners, and integrations offered by cloud service providers continues to expand.

In general, IaaS offerings require the customer to handle more layers of management than PaaS and SaaS options.

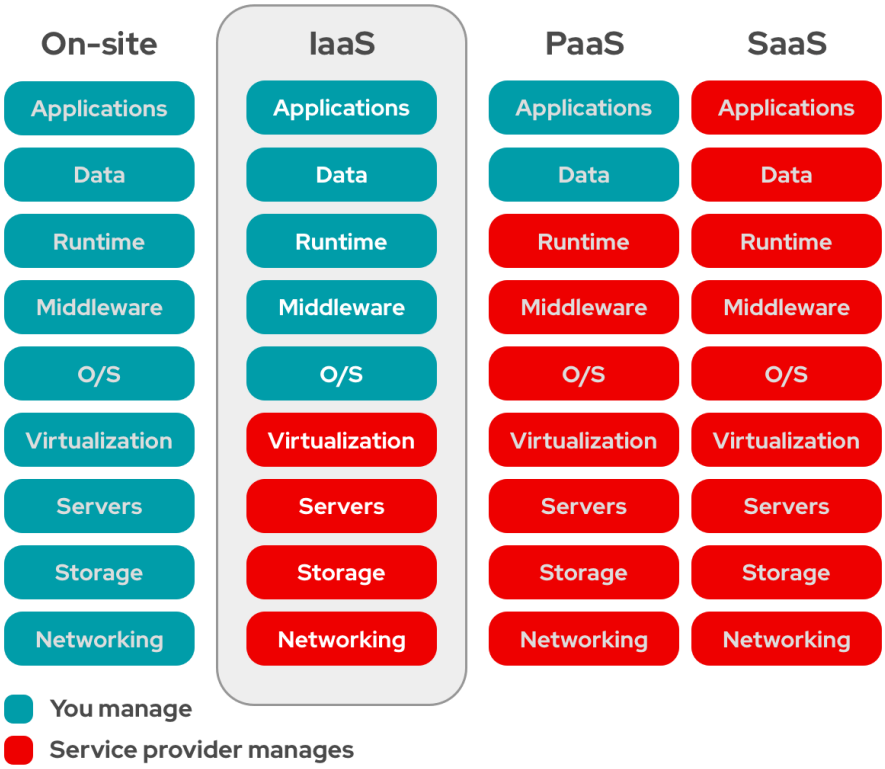


Figure 4:IaaS vs PaaS vs SaaS

With PaaS, hardware and an application-software platform are provided and managed by an outside service provider, but the user handles the actual application and data. Primarily for developers and programmers, PaaS gives the user a platform on which to develop, run, and manage their own applications without having to build and maintain the infrastructure usually associated with the process. Cloud platforms are a type of PaaS, which includes services provided by Amazon Web Services (AWS), Microsoft Azure, Google Cloud, and IBM Cloud.

SaaS is a service that delivers a web application, which the service provider manages, to its users typically through a web browser. Software updates, bug fixes, and other general software maintenance are taken care of for the user, and they connect to the app through a dashboard or API. SaaS also eliminates the need to have an application installed locally on each individual user's computer, allowing greater methods of group or team access to the software as desired.

As-a-service offerings are especially appealing for customers whose workloads have changed, since the service providers can provide the expertise and effort required to handle scalability and load balancing.

- **How IaaS relates to virtualization, automation, and containerization**

An IaaS provider is able to simplify the developer experience by removing the overhead of setting up managing servers. This usually relies on cloud computing architecture supported by virtualization, containers, and automation. To the developer, applications can just be built and deployed, as the server management is abstracted away.

Through virtualization, virtual machines (VMs) provide complete environments that function as a virtual computer system with their own CPUs, memory, network interface, and storage. In IaaS, these are created on a physical hardware system in a data center. Software called a hypervisor separates the machine's resources from the hardware and provisions them appropriately so they can be used by the VM.

IT automation is at work beneath the service of every IaaS offering, enabling the underlying virtual machines and other infrastructure to be deployed seamlessly, and scale up and down as necessary to meet demand. The automation of multiple tasks and configurations across groups of systems or machines is known as orchestration.

An IaaS offering may also offer support for containerization, in which software code with all its necessary components like libraries, frameworks, and other dependencies are packaged in their own Linux container, ready for deployment into a computing environment (which can be a VM). In contrast to virtual machines, containers do not include their own operating system (OS) and thus can be much smaller in size.

Specific IaaS solutions can help developers working with containers. One solution is Kubernetes, an open source container orchestration platform that helps manage distributed, containerized applications at massive scale. Kubernetes takes care of automating the deployment and management of containers. Some IaaS providers offer Kubernetes as a service.

❖ What does IaaS have in common with DevOps?

DevOps describes a way of working at the intersection of development and operations. It emphasizes reducing the time it takes for software improvements to advance deployment, so users gain access to new applications faster. DevOps approaches require development teams and operations teams to communicate frequently and collaborate as teammates.

DevOps goes hand in hand with frequent changes to code and dynamic infrastructure usage, which can make it a good fit for IaaS. DevOps emphasizes automating routine operational tasks and standardizing environments across an app's lifecycle. DevOps teams will often build their software using a microservices architecture and link these services together with APIs. This all helps teams deliver software faster by focusing on creating smaller pieces of functionality, then using strategies such as agile methodology to bring it all together.

IaaS by reducing the need to maintain server infrastructure and emphasizing a simpler, automated developer experience can help support DevOps workflows.

❖ What's the difference between IaaS and serverless?

Serverless computing describes a cloud-native development model in which servers are abstracted away from app development, and it's commonly associated with IaaS.

Serverless relies on a cloud provider to manage both the infrastructure and the scaling of apps. Serverless apps are deployed in containers that automatically launch on demand when called.

Under IaaS, it generally remains the user's responsibility to scale up server capacity during times of high demand and to scale down when that capacity is no longer needed. The cloud infrastructure necessary to run an app is active even when the app isn't being used.

With serverless architecture, by contrast, apps are launched only as needed. When an event triggers app code to run, the public cloud provider dynamically allocates resources for that code. The user stops paying when the code finishes executing. With serverless, routine tasks such as managing the operating system and file system, security patches, load balancing, capacity management, scaling, logging, and monitoring are all offloaded to a cloud services provider.

Things to consider when choosing an IaaS provider

- Flexibility: Purchase only the components you need for your use case and scale them up or down based on business needs.
- Affordability: Low overhead and no maintenance costs make IaaS an affordable option. Pay only for what you use and how often you use it like paying a utility bill.
- Control: The user has control of their infrastructure.
- Security: Does the provider have a trusted reputation and the resources to prevent and manage any security threats? Are there documented disaster recovery protocols that ensure business continuity?
- Multitenant systems: As IaaS providers tend to allocate infrastructure resources to multiple clients as needed, the provider is required to make sure that customers are unable to access each other's data. Having multiple customers using a provider's infrastructure can create an imbalance known as *noisy neighbor*s where a single user's monopoly of a specific resource can slow down performance for others—so providers need to plan resource allocation carefully. This is why it's important to understand how the provider scales with their consumer loads.
- Service: What are the service provider's service-level agreements (SLAs)the minimum amount of time and effort the provider guarantees to solve resource provisioning issues?

- Reliability: Performance and speed largely depend on the provider. Any software or hardware problems on their end will impact user runtimes.

III.3 Service Connectivity

Service connectivity expresses the requirements of a service in terms of transmission bit rates and the locations to be connected. This type of connectivity is sometimes called a *service trial*.

Because service connectivity represents required bandwidth rather than available bandwidth, it cannot have riders. Service connectivity consumes other types of connectivity and resources but cannot be consumed itself. You can enable service connectivities with channelized connectivity, packet connectivity, or both. The bit rate of service connectivity cannot exceed that of its enabling connectivity.

Service connectivities are not defined by technology, so when you create a Service Connectivity specification in Design Studio, the technology is automatically set to Undefined.

Service Connectivity specifications include an attribute (Service Connectivity Controlled) that determines how you create and use service connectivities based on the specification:

- If Service Connectivity Controlled is not enabled, you create and configure the service connectivity normally. Although it is created outside the context of service configurations, you can include the service connectivity in service configurations.
- If Service Connectivity Controlled is enabled, you can create the service connectivity only as part of a service configuration. This type of service connectivity is used in packet-based service arrangements, such as Carrier Ethernet.

❖ Service Connectivity Examples

You can use service connectivities to represent services such as VoIP that are enabled by other connectivity. For example, Figure 1 illustrates a VoIP service that is enabled by an OC192 connectivity in one segment and a 10 Gbps Ethernet connectivity in another segment.

You can also use service connectivities to represent circuits that are designed internally by a provider and are not initially part of a service. They can be included subsequently in customer service designs. These connectivities are sometimes called Official Use or Company Use circuits.

For example, service connectivity could be used to represent an X2 link connecting cell towers in an LTE backhaul scenario. Figure 5 illustrates an X2 link enabled by other connectivities of various technologies.

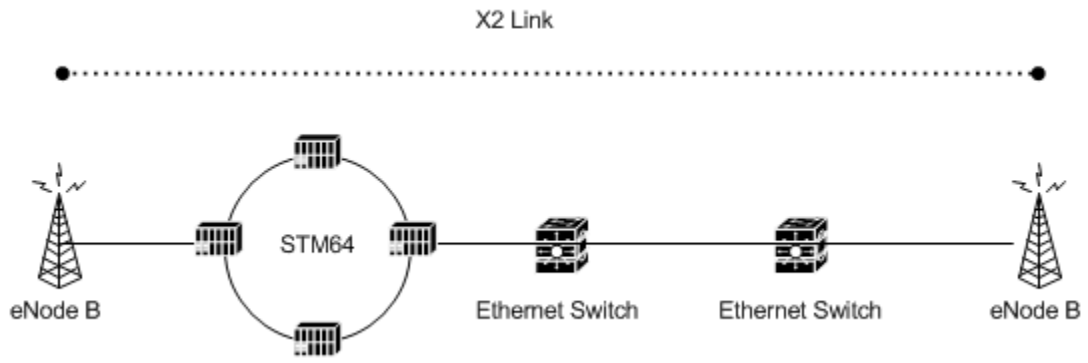


Figure 5:Service Connectivity for LTE Backhaul

❖ Service Configuration Controlled Service Connectivity

Unlike other connectivity entities, you do not create service configuration-controlled connectivities directly. Instead, you create them in the process of configuring the services to which they are assigned. For example, when you design an Ethernet Access service as part of a Carrier Ethernet E-LAN service, you create a service connectivity. You assign the service connectivity as a configuration item in the Ethernet Access service configuration.

The life cycle of a service connectivity is bound to that of the Service configuration to which it is assigned. When the service connectivity is first created, it can be viewed only from within the Service configuration. It is in Pending Install status and cannot be searched for in UIM.

When the Service configuration is completed, the service connectivity is transitioned to the Current context. It is then visible in the search results.

Similarly, when you disconnect a service, the parent service configuration transitions to Pending Disconnect status. This transition propagates to the service connectivity, and it is also transitioned to Pending Disconnect. Completing the disconnect of the service completes the disconnect of the service connectivity.

You use service configuration-controlled service connectivities in two contexts:

- Connecting service locations to the provider network in multipoint services. For example, in a Carrier Ethernet E-LAN service, there are separate Ethernet Access services for each service location. Service connectivity is assigned to the Ethernet Access service configuration.

- Representing end-to-end service connectivity in point-to-point services. For example, in a Carrier Ethernet E-Line service, service connectivity is assigned to the service configuration.

Service connectivity designs are created automatically based on the configuration of their parent service. The Connectivity Design tab for service connectivities is read-only.

This type of service connectivity always has at least one end point at a service location. The other end point can be at another service location or at a network location. When assigned to Ethernet Access services, one end point is at a service location and the other is at a network location. When assigned to a point-to-point service such as an E-Line service, both end points are at service locations.

❖ Service Connectivity in Multipoint Services

In a multipoint Carrier Ethernet network, service connectivities are network segments that connect service locations to a service provider network. For each service location, there is an Ethernet Access Service with corresponding service connectivity. The service connectivity is enabled by the UNI connectivity that connects the CE device at the service location to the PE device at the provider network.

❖ Service Connectivities in Point-to-Point Services

In a point-to-point service, such as an Ethernet E-Line service, the service connectivity spans the whole service from the A service location to the Z service location.

The service connectivity is enabled by UNI connectivities on the service location ends and network transport inside the service provider networks.

A gap enables the service connectivity within the provider network because the service connectivity reflects only the UNI connectivities up to their PVN trunk interfaces. The gap is automatically labeled Gap – Network Transport.

Chapter IV: Technical Project

In this chapter, I will present the project I worked on during my internship. I will begin by providing an overview of the main technologies involved, followed by a detailed discussion of the networking concepts applied throughout the project and I will explain the configurations implemented as they become relevant in each section.

IV.1 Project Description

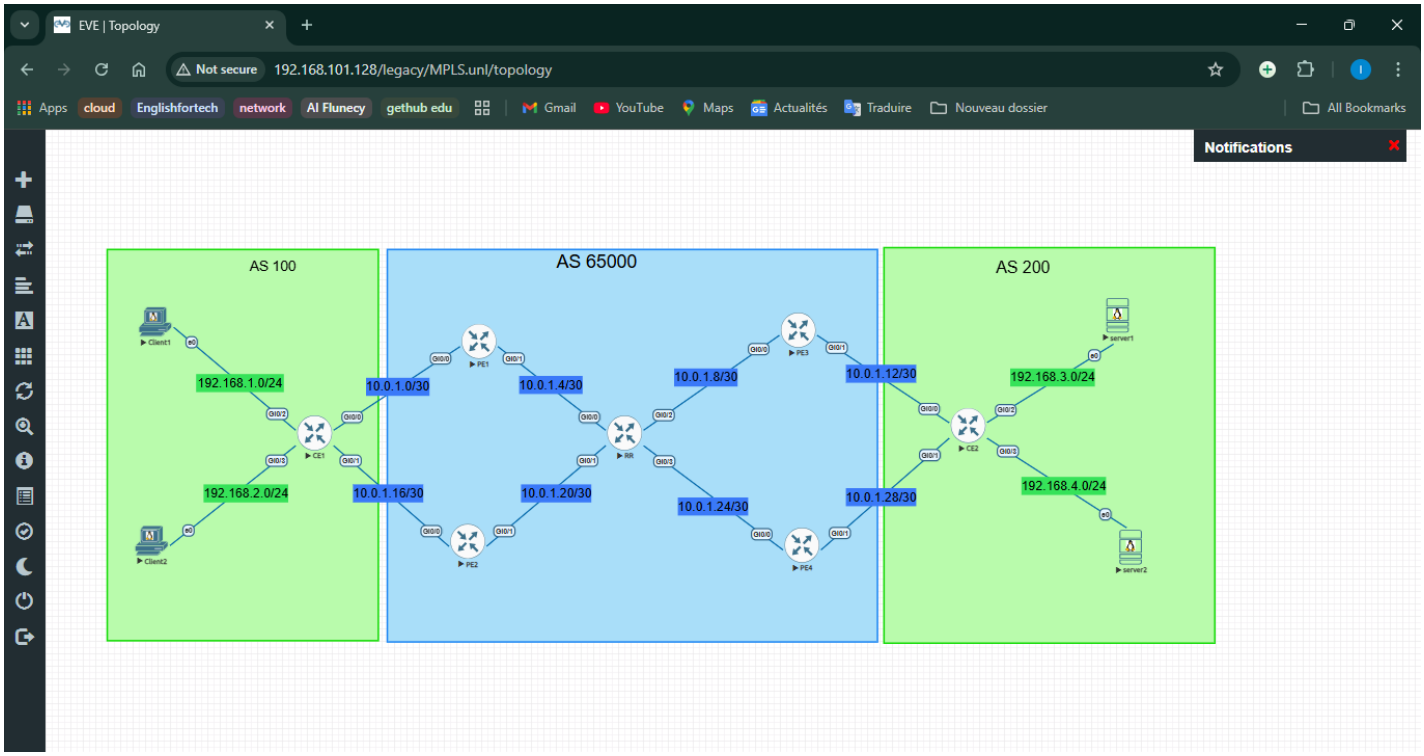


Figure 6:Overview of the topology

❖ Network Topology Description

The topology consists of three Autonomous Systems (ASs):

- AS 100 – Client domain
- AS 65000 – Backbone (Service Provider core)
- AS 200 – Server domain

❖ **Configuration Details**

- The Backbone (AS 65000) uses OSPF as its Interior Gateway Protocol (IGP), operating within a single area.

- In addition to the physical interconnection IP addresses, each router is assigned a logical Loopback interface (/32), which is also advertised in the OSPF process.
- PE1, PE2, PE3, and PE4 rely on a Route Reflector (RR) for iBGP route exchange within AS 65000.
- Loopback addresses are used as iBGP peering sources.
- CE1 and CE2 establish eBGP sessions with their respective Provider Edge (PE) routers to advertise Client and Server routes.

❖ **Expected Results**

1. OSPF, iBGP, and eBGP sessions must be properly established.

- CE1 should receive the server routes.
- CE2 should receive the client routes.

2. Routing tables on the necessary devices must be adjusted to enforce distinct traffic paths between each client and its corresponding server:

- Client 1 ↔ CE1 ↔ PE1 ↔ RR ↔ PE3 ↔ CE2 ↔ Server 1
- Client 2 ↔ CE1 ↔ PE2 ↔ RR ↔ PE4 ↔ CE2 ↔ Server 2

IV.2 Introduction to EVE-NG

Figure 7:EVE-NG logo

❖ What is EVE-NG?

To describe what Emulated Virtual Environment – Next Generation (EVE-NG) is without solely stating dry facts about features, we need to elaborate more on what EVE-NG can be used for and whom it would be useful for. In some trivial dry words, EVE-NG gives you tools to use

around virtual devices and interconnect them with other virtual or physical devices. Many of its features greatly simplify the usability, re-usability, manageability, interconnectivity, distribution and therefore the ability to understand and share topologies, work, ideas, concepts or simply "labs". This can simply mean it will reduce the cost and time to set up what you need, or it might enable you to do tasks you would not have thought could be done this simply.

❖ What is EVE-NG used for?

This is the real question but there is no finite answer, the possibilities are almost limitless and depends on what you want to use it for. It can be used for studying all kinds of technologies. You can learn about general technologies or vendor specific topics. You can test new technologies like network automation, SDN, etc. It can be used to create corporate networks and test changes before putting them into production. You can create proof of concepts for clients. You can troubleshoot network issues by recreating them and e.g. use Wireshark to inspect packets. It is most definitely not just for networking, it can be used to test software in simulated networks, test out security vulnerabilities of any kind, system engineering like LDAP and AD servers and many more areas. You could set it up to automate sandboxing unknown files/software and use software to analyze short- and long-term behavior for malicious intent much simpler than without EVE-NG. The list of what EVE-NG can be used for could go on indefinitely, possibilities are limited by knowledge and imagination only

IV.3 Introduction to VMware Workstation

Figure 8:VMware workstation logo

❖ What is VMware workstation

VMware Workstation is software (hosted hypervisor) that makes it easy to create and manage virtual machines on Windows (64-bit only) and Linux. On a single computer, multiple virtual machines can run concurrently with a PC, and each virtual machine can run a different operating system (such as BSD, Linux, Windows, or MS-DOS) at the user's option.

You can use either the premium version of VMware Workstation Pro or the unpaid version of VMware Workstation Player (for non-commercial usage) of VMware Workstation. This hypervisor allows a virtual machine to share both physical and portable disks. The user can take snapshots, restore the virtual computer to the preferred condition, and access extra services with the commercial version, if necessary.

In addition, users also create inventory folders that can contain multiple virtual machines. They can be managed as a single mechanism useful for testing in a variety of environments.

❖ What is VMware for?

1. Learn Server Management and Web Development

One of the earliest use cases for VMware Workstation is learning server-side web development. You can visit the VMware Solution Exchange website and find the right device for your needs.

2. Testing Multiple Operating Systems and Applications

VMware Workstation allows you to run operating systems other than those currently installed on your computer. This way, you can easily test new features in Windows Server 2012 or conceptualize Skype for Business and more.

3. The IT Administrator's Best Friend

Using a virtual machine, you can test and validate all configuration changes on a legacy system before deploying to a production environment. This way, IT administrators can easily prove to management that a new application works. At the same time, an IT administrator can use an almost unlimited number of VMware workstations for his own purposes - this is great.

4. Product Demos

With VMware Workstation, system administrators can run application demos on potential customer sites anytime, anywhere.

5. vSphere Staging

With vSphere, you can move virtual machines from your production environment to your workstation on a virtualized private network. This is required for various updates and configuration changes before returning them to the vSphere production environment.

6. Software Development

To develop any software, it is important to have the right environment for both building and testing your application. Indeed, with the help of VMware Workstation, software developers can test an application on several operating systems, take pictures of virtual machines both before and after installing their application.

7. Browser Testing

For JavaScript applications that run in a browser, testing is an essential component. For example, you will see that a web application works fine using Google Chrome in Windows 7, but errors may be found in the same browser, but in Linux or OS X. So, in order to test several browsers and versions in several operating systems, you need virtual machines.

❖ How to set up VMware workstation

1. System Requirements

- 64-bit CPU with Intel VT-x or AMD-V enabled in BIOS/UEFI.
- Minimum 4 GB RAM (8–16 GB recommended).
- At least 1 GB for installation + 20–60 GB per VM on SSD.
- Supported 64-bit Windows (10/11) or Linux distributions.
- Ensure Hyper-V is disabled on Windows hosts.

2. Download & Installation

- Download VMware Workstation Pro or Player from the official VMware website.
- On Windows: Run the .exe → Accept license → Typical setup → Finish and reboot.
- On Linux: Run `chmod +x VMware-Workstation.bundle` → `sudo ./VMware-Workstation.bundle`.
- Install VMware Tools after the first VM setup for better integration.

3.Virtual Machine Creation • Open VMware → 'Create a New Virtual Machine'.

- Choose Typical setup → Select ISO → Choose OS type.
- Allocate CPU (2–4 cores) and RAM (2–8 GB). • Create a 40–60 GB virtual disk (split or single file).
- Finish and power on to install the guest OS. 4. Network Configuration
- NAT: VM uses host IP for internet access.
- Bridged: VM connects directly to the local network.
- Host-Only: Isolated network between host and VM.
- Change mode via VM → Settings → Network Adapter → Select mode.

4.Best Practices • Do not overcommit CPU/RAM – leave resources for host.

• Use SSD storage for performance.
• Take snapshots before major changes.
• Regularly update VMware Tools and the host system.

❖ EVE-NG Setup on VMware

1.System Requirements

- CPU: 64-bit Intel/AMD with virtualization support (VT-x/AMD-V enabled). • RAM: Minimum 8 GB (16 GB+ recommended)
- Storage: At least 80 GB free (SSD preferred).
- Host OS: Windows 10/11 or Linux (64-bit).
- VMware Workstation Pro or Player installed.

2.Download & Installation

- Go to https://www.eve-ng.net → Download the EVE-NG Community OVA file.
- Open VMware Workstation → File → Open → Select the EVE-NG .ova file.
- Import the appliance → Review VM settings → Finish.
- Edit VM settings: Assign 8+ GB RAM, 4+ vCPUs, and Bridged networking mode.
- Power on the VM → EVE-NG will boot and display a management IP address.

3.Configuration

- Access EVE-NG Web UI: Open a browser → Enter displayed IP (e.g., https://192.168.x.x).
- Login with default credentials: Username: admin | Password: eve.
- Upload device images to /opt/unetlab/addons/qemu/ using WinSCP or FileZilla (in my case I used WinSCP).

Figure 10:FileZilla logo

Figure 9: WinSCP

- Fix permissions by running `/opt/unetlab/wrappers/unl_wrapper -a fixpermissions` via SSH.
- Create a new lab: New → Name → Add nodes (routers, switches, PCs).

4.Verification

Figure 11: Ultra VNC logo

Figure 12: PuTTY logo

- Open EVE-NG Web UI → Start lab.

- Power on all nodes → Check console access via Telnet/SSH (in my case I used Putty and UltraVNC)
- Ping between nodes to confirm connectivity.
- Ensure EVE-NG dashboard displays 'All nodes running'.

IV.4 IP Routing

3.1 Definition and Role

IP Routing is the fundamental process by which routers determine the optimal path for forwarding packets from a source to a destination based on their IP addresses.

Each router maintains a routing table that lists reachable networks and the next hop toward them.

In this project, static routing was not used; instead, dynamic routing protocols (such as DHCP, OSPF and BGP) handled route exchange. However, basic IP addressing and CEF (Cisco Express Forwarding) were configured to ensure efficient packet forwarding.

3.2 Implementation

All routers in the topology use IP routes on their interfaces. Examples:

CE1

```
ip dhcp pool client1-pool
 network 192.168.1.0 255.255.255.0
 default-router 192.168.1.1
 dns-server 8.8.8.8
!
ip dhcp pool client2-pool
 network 192.168.2.0 255.255.255.0
 default-router 192.168.2.1
 dns-server 8.8.8.8
!
ip cef
no ipv6 cef
!
multilink bundle-name authenticated
!
redundancy
!
```

```
interface Loopback0
 ip address 11.11.11.11 255.255.255.255
!
interface GigabitEthernet0/0
 ip address 10.0.1.1 255.255.255.252
 duplex auto
 speed auto
 media-type rj45
!
interface GigabitEthernet0/1
 ip address 10.0.1.17 255.255.255.252
 duplex auto
 speed auto
 media-type rj45
!
interface GigabitEthernet0/2
 ip address 192.168.1.1 255.255.255.0
 duplex auto
 speed auto
 media-type rj45
!
interface GigabitEthernet0/3
 ip address 192.168.2.1 255.255.255.0
 duplex auto
 speed auto
 media-type rj45
```

PE1

```
interface Loopback0
 ip address 1.1.1.1 255.255.255.255
 ip ospf 110 area 0
!
interface GigabitEthernet0/0
 ip address 10.0.1.2 255.255.255.252
 duplex auto
 speed auto
 media-type rj45
!
interface GigabitEthernet0/1
 ip address 10.0.1.5 255.255.255.252
 ip ospf 110 area 0
 duplex auto
 speed auto
 media-type rj45
 mpls ip
!
interface GigabitEthernet0/2
 no ip address
 shutdown
 duplex auto
 speed auto
 media-type rj45
!
```

```
interface GigabitEthernet0/3
 no ip address
 shutdown
 duplex auto
 speed auto
 media-type rj45
```

RR

```
interface Loopback0
 ip address 5.5.5.5 255.255.255.255
 ip ospf 110 area 0
!
interface GigabitEthernet0/0
 ip address 10.0.1.6 255.255.255.252
 ip ospf 110 area 0
 duplex auto
 speed auto
 media-type rj45
 mpls ip
!
interface GigabitEthernet0/1
 ip address 10.0.1.22 255.255.255.252
 ip ospf 110 area 0
 duplex auto
 speed auto
 media-type rj45
 mpls ip
```

```
!
interface GigabitEthernet0/2
 ip address 10.0.1.9 255.255.255.252
 ip ospf 110 area 0
 duplex auto
 speed auto
 media-type rj45
 mpls ip
!
interface GigabitEthernet0/3
 ip address 10.0.1.25 255.255.255.252
 ip ospf 110 area 0
 duplex auto
 speed auto
 media-type rj45
 mpls ip
```

All routers also have:

```
 ip cef
```

which enables Cisco Express Forwarding, improving performance by building a hardware-optimized forwarding table.

3.3 Analysis

The base IP configuration ensures **connectivity and reachability** between routers. Combined with CEF, it guarantees optimal forwarding performance, serving as the backbone for higher-level routing protocols like OSPF and BGP.

IV.5 OSPF

5.1 Definition and Role

OSPF is a link-state Interior Gateway Protocol (IGP) that dynamically exchanges routing information within a single autonomous system.
It divides networks into areas, where Area 0 (Backbone) is essential for interconnecting other areas. OSPF uses Dijkstra's algorithm to determine the shortest path to each destination.

In this topology, OSPF was implemented within AS 65000 (the service provider core) to provide intra-domain connectivity and loopback reachability for BGP and MPLS.

5.2 Implementation

Example: **PE1**

```
interface Loopback0
ip ospf 110 area 0
interface GigabitEthernet0/1
 ip ospf 110 area 0
router ospf 110
 network 1.1.1.1 0.0.0.0 area 0
 network 10.0.1.0 0.0.0.255 area 0
```

Example: RR

```
router ospf 110
 network 5.5.5.5 0.0.0.0 area 0
 network 10.0.1.0 0.0.0.255 area 0
```

5.3 Analysis

All PE routers and the Route Reflector participate in OSPF Area 0, forming a single connected backbone.
Each router advertises its loopback interface, ensuring that all PEs and the RR can reach one another a prerequisite for iBGP sessions and MPLS label distribution.

IV.6 MPLS

6.1 Definition and Role

MPLS enhances packet forwarding by attaching labels to packets, enabling routers (Label Switch Routers) to forward traffic based on labels instead of IP lookups.
It combines the efficiency of Layer 2 switching with Layer 3 routing intelligence, forming the foundation of many Service Provider networks.

In this project, MPLS was enabled across the core network (AS 65000), creating Label Switched Paths (LSPs) between all Provider Edge routers.

6.2 Implementation

Example: PE1

```
interface GigabitEthernet0/1
 ip ospf 110 area 0
 mpls ip
```

Example: RR

```
interface GigabitEthernet0/0
 ip ospf 110 area 0
 mpls ip
interface GigabitEthernet0/1
 ip ospf 110 area 0
 mpls ip
```

6.3 Analysis

MPLS provides the transport layer for inter-site communication. By enabling mpls ip on all backbone-facing interfaces, label-based forwarding is established across the core.
This ensures efficient traffic flow and sets the foundation for advanced services like VPNs and traffic engineering.

IV.7 iBGP and eBGP

7.1 Definition and Role

BGP (Border Gateway Protocol) is the primary protocol for exchanging routing information between autonomous systems.
Two forms exist:

- eBGP: Exchanges routes between different ASes (e.g., CE–PE communication).
- iBGP: Shares routes internally within one AS (e.g., between PEs through the RR).

In large networks, full iBGP meshes become complex. To simplify, a Route Reflector (RR) was implemented in AS 65000 to manage route distribution.

7.2 Implementation

iBGP : PE to RR

```
router bgp 65000
 neighbor 5.5.5.5 remote-as 65000
 neighbor 5.5.5.5 update-source Loopback0
 address-family ipv4
  neighbor 5.5.5.5 activate
  neighbor 5.5.5.5 next-hop-self
 exit-address-family
```

RR

```
router bgp 65000
 neighbor 1.1.1.1 remote-as 65000
```

```
 neighbor 1.1.1.1 route-reflector-client
 neighbor 2.2.2.2 route-reflector-client
 neighbor 3.3.3.3 route-reflector-client
 neighbor 4.4.4.4 route-reflector-client
```

eBGP: CE1 to PE1

```
router bgp 100
 neighbor 10.0.1.2 remote-as 65000
 address-family ipv4
  network 192.168.1.0
  network 192.168.2.0
```

7.3 Analysis

The combination of iBGP and eBGP ensures end-to-end route propagation between customer and provider networks. The Route Reflector eliminates the need for full-mesh peering, while loopback-based sessions improve reliability. This configuration models real-world ISP backbones, where scalability and redundancy are essential.

IV.8 BGP Path Control

8.1 Definition and Role

BGP Path Control allows administrators to influence route selection using attributes such as Local Preference and AS Path Prepending.
This enables traffic engineering determining which links are preferred for outgoing or incoming traffic without physically altering the network.

8.2 Implementation

PE1 Example:

```
route-map TAG-client1 permit 20
 match ip address prefix-list client1
 set local-preference 200
route-map TAG-server permit 22
 match ip address prefix-list server2
```

```
 set as-path prepend 65000 65000
```

PE3 Example

```
route-map TAG-server1 permit 10
 match ip address prefix-list server1
 set local-preference 200
```

PE4 Example

```
route-map TAG-server2 permit 10
 match ip address prefix-list server2
 set local-preference 200
```

8.3 Analysis

By assigning a higher local preference, routers prefer specific outbound paths:

- Client1 traffic → PE1 → PE3 → Server1
- Client2 traffic → PE2 → PE4 → Server2

When the preferred path fails, AS-path prepending allows BGP to automatically choose the backup route.
This method provides redundancy and load balancing, achieving intelligent and policy-driven routing control.

IV.9 Testing and Verification

Now that we have completed the configurations, we will move on to the different verification steps performed to ensure the correct operation of the IP, IGP, BGP, MPLS, and path-control configurations implemented across the service provider topology. Testing was conducted progressively, validating each layer of the network before introducing the next protocol. This structured approach ensured stability, simplified troubleshooting, and allowed clear isolation of potential misconfigurations.

9.1 Connectivity Tests

The first phase of testing focused on basic IP reachability. All core routers (PEs and RR) use loopback interfaces as BGP identifiers and as MPLS LDP router-IDs.
Verifying end-to-end loopback reachability confirmed that:

- ✓ OSPF adjacencies were correctly established.
- ✓ The IGP learned all internal subnets.
- ✓ Next-hop resolution for BGP and MPLS would function correctly.

Example:

- ❖ From PE1 (loopback 1.1.1.1) ping 4.4.4.4 (PE4 loopback)

```
PE1#ping 4.4.4.4
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 4.4.4.4, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/2/4 ms
PE1#
```

Figure 13: Ping test 1

- ❖ From PE1 (loopback 1.1.1.1) ping 5.5.5.5 (RR loopback)

```
PE1#ping 5.5.5.5
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 5.5.5.5, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/3 ms
PE1#
```

Figure 14: Ping test 2

- ❖ From PE3 (loopback 3.3.3.3) ping 4.4.4.4 (PE4 loopback)

```
PE3#ping 4.4.4.4
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 4.4.4.4, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/2/4 ms
PE3#
```

Figure 15: Ping test 3

- ✓ Conclusion: the test has been successful; loopback reachability is functioning.

9.2 OSPF Verification

To validate OSPF we should ensure that all PEs and the RR had built adjacencies and exchanged LSAs correctly. To do that we will need to check OSPF neighboring in RR, if the RR

is aware of his OSPF neighbors, that will mean that OSPF is successfully implemented on all the PEs.

```
RR#sh ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
4.4.4.4           1   FULL/BDR        00:00:35    10.0.1.26       GigabitEthernet0/3
3.3.3.3           1   FULL/BDR        00:00:32    10.0.1.10       GigabitEthernet0/2
2.2.2.2           1   FULL/BDR        00:00:33    10.0.1.21       GigabitEthernet0/1
1.1.1.1           1   FULL/BDR        00:00:33    10.0.1.5        GigabitEthernet0/0
RR#
```

Figure 16: OSPF neighbors

```
RR#sh ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      1.0.0.0/32 is subnetted, 1 subnets
O        1.1.1.1 [110/2] via 10.0.1.5, 02:11:13, GigabitEthernet0/0
      2.0.0.0/32 is subnetted, 1 subnets
O        2.2.2.2 [110/2] via 10.0.1.21, 02:11:13, GigabitEthernet0/1
      3.0.0.0/32 is subnetted, 1 subnets
O        3.3.3.3 [110/2] via 10.0.1.10, 02:11:03, GigabitEthernet0/2
      4.0.0.0/32 is subnetted, 1 subnets
O        4.4.4.4 [110/2] via 10.0.1.26, 02:11:13, GigabitEthernet0/3
      10.0.0.0/8 is variably subnetted, 12 subnets, 2 masks
O        10.0.1.0/30 [110/2] via 10.0.1.5, 02:11:13, GigabitEthernet0/0
O        10.0.1.12/30 [110/2] via 10.0.1.10, 02:11:03, GigabitEthernet0/2
O        10.0.1.16/30 [110/2] via 10.0.1.21, 02:11:13, GigabitEthernet0/1
O        10.0.1.28/30 [110/2] via 10.0.1.26, 02:11:13, GigabitEthernet0/3
RR#
```

Figure 17: OSPF routes

```
RR#show ip ospf database

            OSPF Router with ID (5.5.5.5) (Process ID 110)

                Router Link States (Area 0)

Link ID         ADV Router      Age         Seq#       Checksum Link count
1.1.1.1         1.1.1.1         1930        0x80000007 0x00D6FE 3
2.2.2.2         2.2.2.2         1962        0x80000007 0x009207 3
3.3.3.3         3.3.3.3         1969        0x80000007 0x00B0F8 3
4.4.4.4         4.4.4.4         1958        0x80000007 0x006C01 3
5.5.5.5         5.5.5.5         1946        0x80000009 0x009A3E 5

                Net Link States (Area 0)

Link ID         ADV Router      Age         Seq#       Checksum
10.0.1.6        5.5.5.5         1946        0x80000004 0x0004F8
10.0.1.9        5.5.5.5         1946        0x80000004 0x004AA7
10.0.1.22       5.5.5.5         1946        0x80000004 0x009553
10.0.1.25       5.5.5.5         1946        0x80000004 0x00DB02
RR#
RR#
RR#
```

Figure 18: OSPF database

- ✓ Conclusion: OSPF has been successfully implemented, and the routes have been successfully exchanged between routers.

9.3 iBGP and Route Reflector Verification

Now that we have validated our IGP, BGP sessions must be tested to ensure:

- ✓ All PEs successfully formed iBGP sessions with the Route Reflector.
- ✓ Routes were correctly reflected between clients.
- ✓ Route-maps and prefix-lists were applied correctly.

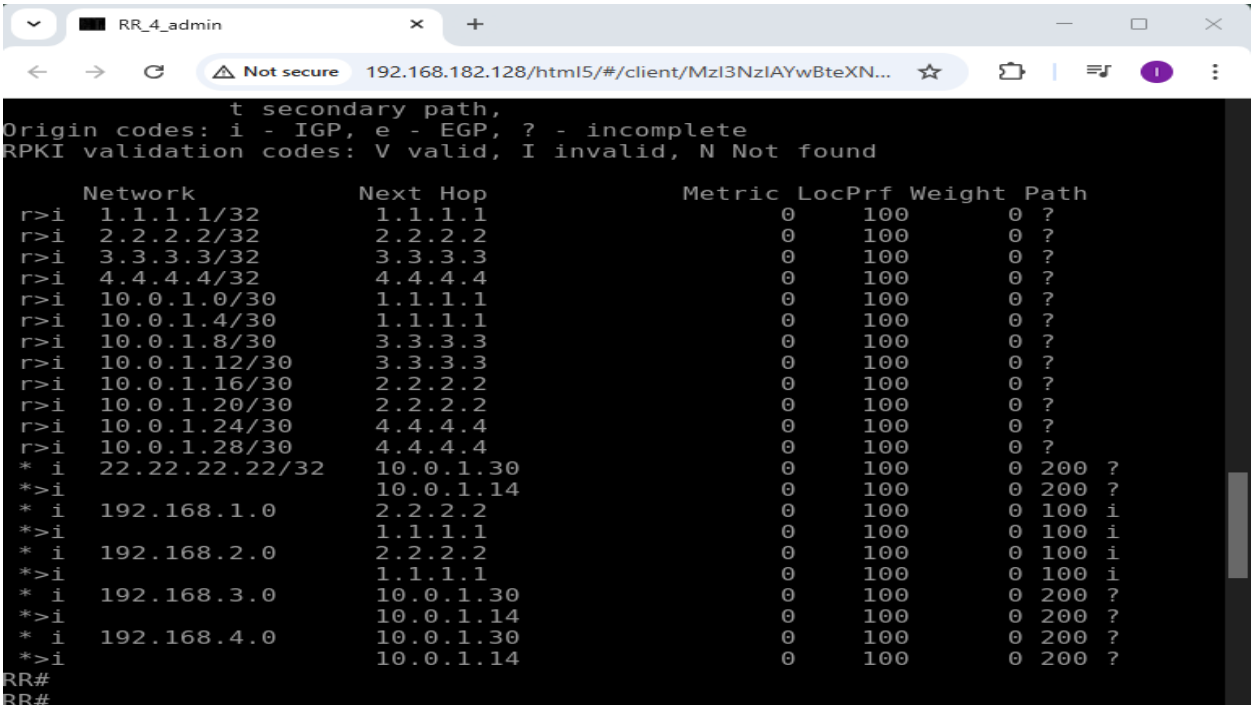
BGP table with Route-map and prefix-list applied:

```
RR#sh ip bgp
BGP table version is 5, local router ID is 5.5.5.5
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
              t secondary path,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>i  192.168.1.0      1.1.1.1                  0    200      0 100 i
 *>i  192.168.2.0      2.2.2.2                  0    200      0 100 i
 *>i  192.168.3.0      10.0.1.14                0    200      0 200 i
 *>i  192.168.4.0      10.0.1.30                0    200      0 200 i
RR#
```

Figure 19: BGP routing table 1

BGP table without Route-map and prefix-list applied:



```
                  t secondary path,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 r>i  1.1.1.1/32       1.1.1.1                  0    100      0 ?
 r>i  2.2.2.2/32       2.2.2.2                  0    100      0 ?
 r>i  3.3.3.3/32       3.3.3.3                  0    100      0 ?
 r>i  4.4.4.4/32       4.4.4.4                  0    100      0 ?
 r>i  10.0.1.0/30      1.1.1.1                  0    100      0 ?
 r>i  10.0.1.4/30      1.1.1.1                  0    100      0 ?
 r>i  10.0.1.8/30      3.3.3.3                  0    100      0 ?
 r>i  10.0.1.12/30     3.3.3.3                  0    100      0 ?
 r>i  10.0.1.16/30     2.2.2.2                  0    100      0 ?
 r>i  10.0.1.20/30     2.2.2.2                  0    100      0 ?
 r>i  10.0.1.24/30     4.4.4.4                  0    100      0 ?
 r>i  10.0.1.28/30     4.4.4.4                  0    100      0 ?
 * i  22.22.22.22/32   10.0.1.30                0    100      0 200 ?
 *>i                   10.0.1.14                0    100      0 200 ?
 * i  192.168.1.0      2.2.2.2                  0    100      0 100 i
 *>i                   1.1.1.1                  0    100      0 100 i
 * i  192.168.2.0      2.2.2.2                  0    100      0 100 i
 *>i                   1.1.1.1                  0    100      0 100 i
 * i  192.168.3.0      10.0.1.30                0    100      0 200 ?
 *>i                   10.0.1.14                0    100      0 200 ?
 * i  192.168.4.0      10.0.1.30                0    100      0 200 ?
 *>i                   10.0.1.14                0    100      0 200 ?
RR#
RR#
```

Figure 20: BGP routing table 2

- ✓ Conclusion: The Route Reflector listed each PE as a neighbor in Established state. Each PE receives client and server prefixes according to the path-control policy. Route-map and prefix-list applied can help not just with path control but also in BGP table cleaning.

9.4 MPLS and LDP Verification

BGP is operational, now let's verify MPLS forwarding functionality.
These tests ensure that:

- ✓ LDP sessions were formed.
- ✓ Label bindings were correctly exchanged.

- ✓ Label-switching paths were active.

```
RR#show mpls ldp neighbor
    Peer LDP Ident: 2.2.2.2:0; Local LDP Ident 5.5.5.5:0
        TCP connection: 2.2.2.2.646 - 5.5.5.5.13550
        State: Oper; Msgs sent/rcvd: 201/201; Downstream
        Up time: 02:42:49
        LDP discovery sources:
          GigabitEthernet0/1, Src IP addr: 10.0.1.21
        Addresses bound to peer LDP Ident:
          10.0.1.18       10.0.1.21       2.2.2.2
    Peer LDP Ident: 3.3.3.3:0; Local LDP Ident 5.5.5.5:0
        TCP connection: 3.3.3.3.646 - 5.5.5.5.65388
        State: Oper; Msgs sent/rcvd: 202/202; Downstream
        Up time: 02:42:43
        LDP discovery sources:
          GigabitEthernet0/2, Src IP addr: 10.0.1.10
        Addresses bound to peer LDP Ident:
          10.0.1.10       3.3.3.3         10.0.1.13
    Peer LDP Ident: 4.4.4.4:0; Local LDP Ident 5.5.5.5:0
        TCP connection: 4.4.4.4.646 - 5.5.5.5.32939
        State: Oper; Msgs sent/rcvd: 201/200; Downstream
        Up time: 02:42:42
        LDP discovery sources:
          GigabitEthernet0/3, Src IP addr: 10.0.1.26
        Addresses bound to peer LDP Ident:
          10.0.1.26       4.4.4.4         10.0.1.29
    Peer LDP Ident: 1.1.1.1:0; Local LDP Ident 5.5.5.5:0
        TCP connection: 1.1.1.1.646 - 5.5.5.5.32211
        State: Oper; Msgs sent/rcvd: 202/202; Downstream
        Up time: 02:42:40
        LDP discovery sources:
          GigabitEthernet0/0, Src IP addr: 10.0.1.5
        Addresses bound to peer LDP Ident:
          10.0.1.2        10.0.1.5        1.1.1.1
RR#
```

Figure 21: MPLS forwarding table

```
RR#show mpls forwarding-table
Local      Outgoing   Prefix           Bytes Label   Outgoing   Next Hop
Label      Label      or Tunnel Id     Switched      interface
20         Pop Label  4.4.4.4/32       2850          Gi0/3      10.0.1.26
21         Pop Label  2.2.2.2/32       0             Gi0/1      10.0.1.21
22         Pop Label  1.1.1.1/32       0             Gi0/0      10.0.1.5
23         Pop Label  10.0.1.28/30     0             Gi0/3      10.0.1.26
24         Pop Label  10.0.1.16/30     0             Gi0/1      10.0.1.21
25         Pop Label  10.0.1.0/30      0             Gi0/0      10.0.1.5
26         Pop Label  3.3.3.3/32       0             Gi0/2      10.0.1.10
27         Pop Label  10.0.1.12/30     0             Gi0/2      10.0.1.10
RR#
```

Figure 22: MPLS LDP neighbors

- ✓ Conclusion: All PEs and the RR appeared as LDP neighbors, and MPLS labels are assigned to each prefix learned through IGP.

9.4 Path-Control Verification (Client-to-Server Preferred Paths):

This was one of the key objectives of the project.
The verification confirmed that:

- ✓ Client 1 reached Server 1 via PE1 → RR → PE3
- ✓ Client 2 reached Server 2 via PE2 → RR → PE4
- ✓ Backup paths were available when the primary routes failed
- ✓ AS-path prepending and Local Preference were applied correctly

Check the chosen next-hop at the CE1:

```
Router#sh ip bgp
BGP table version is 12, local router ID is 22.22.22.22
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
              t secondary path,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>   192.168.1.0      10.0.1.13                              0 65000 100 i
 *                     10.0.1.29                              0 65000 65000 6500
0 100 i
 *    192.168.2.0      10.0.1.13                              0 65000 65000 6500
0 100 i
 *>                    10.0.1.29                              0 65000 100 i
 *>   192.168.3.0      0.0.0.0                  0         32768 i
 *>   192.168.4.0      0.0.0.0                  0         32768 i
Router#
```

Check the chosen next-hop at the CE2:

```
CE1#sh ip bgp
BGP table version is 5, local router ID is 11.11.11.11
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
              t secondary path,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>   192.168.1.0      0.0.0.0                  0         32768 i
 *>   192.168.2.0      0.0.0.0                  0         32768 i
 *    192.168.3.0      10.0.1.18                              0 65000 65000 65000 200 i
 *>                    10.0.1.2                               0 65000 200 i
 *>   192.168.4.0      10.0.1.18                              0 65000 200 i
 *                     10.0.1.2                               0 65000 65000 65000 200 i
```

- ✓ Conclusion: Path control has been successfully implemented, and the backup paths are available.

9.5 End-to-End connectivity Tests:

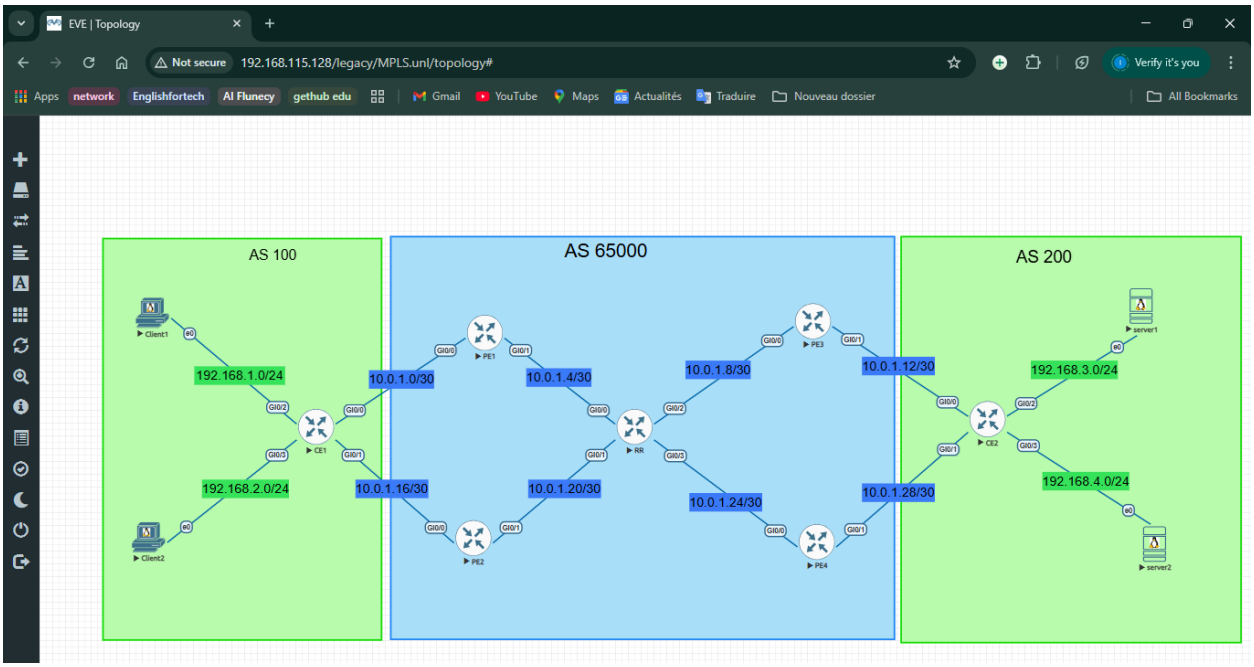
Finally, let's see if the clients and servers can talk to each other.

1. Client–Server ping tests:

```
QEMU (Client1)
root@ubuntu:~#
root@ubuntu:~#
root@ubuntu:~#
root@ubuntu:~# ping 192.168.3.2
PING 192.168.3.2 (192.168.3.2) 56(84) bytes of data.
64 bytes from 192.168.3.2: icmp_seq=1 ttl=59 time=25.3 ms
64 bytes from 192.168.3.2: icmp_seq=2 ttl=59 time=26.6 ms
64 bytes from 192.168.3.2: icmp_seq=3 ttl=59 time=16.9 ms
64 bytes from 192.168.3.2: icmp_seq=4 ttl=59 time=32.4 ms
64 bytes from 192.168.3.2: icmp_seq=5 ttl=59 time=41.1 ms
64 bytes from 192.168.3.2: icmp_seq=6 ttl=59 time=23.6 ms
64 bytes from 192.168.3.2: icmp_seq=7 ttl=59 time=20.6 ms
64 bytes from 192.168.3.2: icmp_seq=8 ttl=59 time=28.8 ms
64 bytes from 192.168.3.2: icmp_seq=9 ttl=59 time=25.1 ms
^C
--- 192.168.3.2 ping statistics ---
9 packets transmitted, 9 received, 0% packet loss, time 8016ms
rtt min/avg/max/mdev = 16.993/26.770/41.101/6.570 ms
root@ubuntu:~#
```

Figure 23: Ping test 4

2. Traceroute to confirm the path taken:



❖ From client 1 traceroute to server 1:

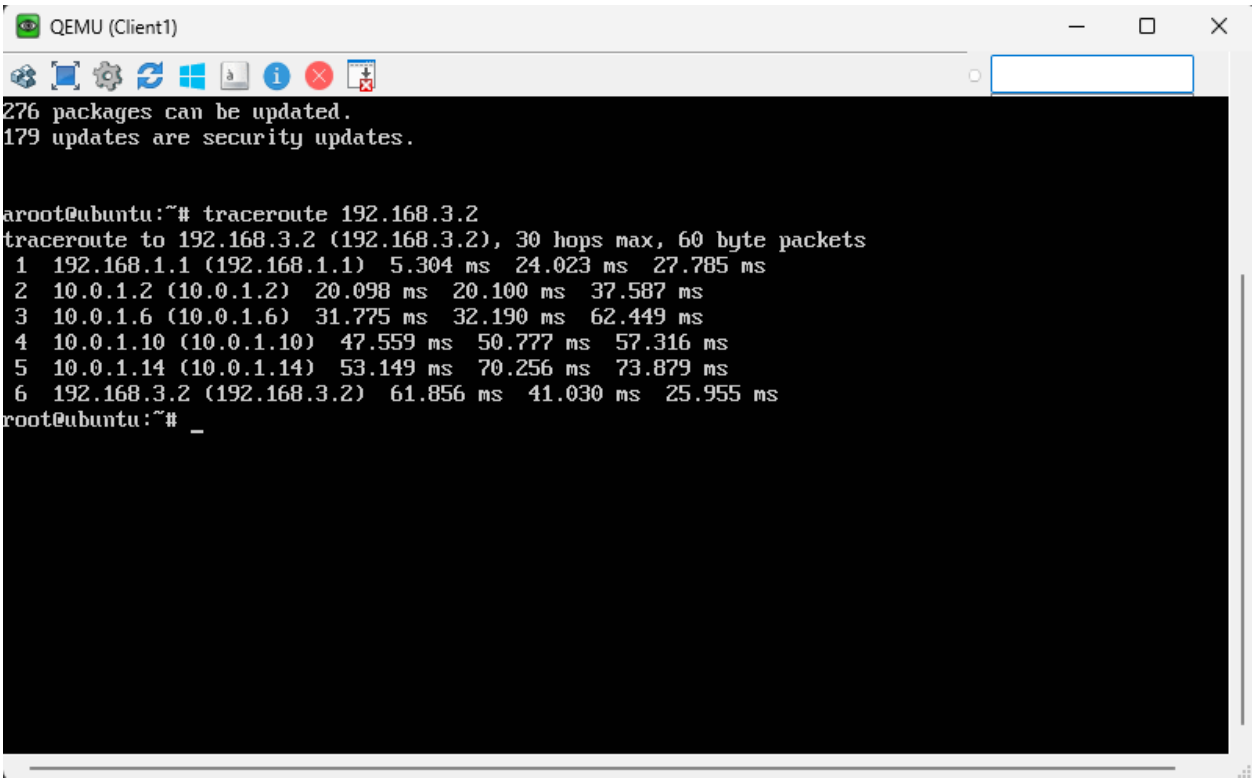


Figure 24: traceroute test 1

- ✓ As expected, the preferred route for client 1 to talk with server 1 is:
  Client 1 ↔ CE1 ↔ PE1 ↔ RR ↔ PE3 ↔ CE2 ↔ Server 1

❖ From client 2 traceroute to server 2:

- ✓ As expected, the preferred route for client 2 to talk with server 2 is:
  Client 2 ↔ CE1 ↔ PE2 ↔ RR ↔ PE4 ↔ CE2 ↔ Server 2

9.6 Failure and Redundancy Testing :

To test redundancy, I will shut down PE1 and PE2 and enter the traceroute command in client 1 to server 1 how CE1 is going to react.

- ✓ After finding out that the preferred route is not available CE1 looks for other possible routes. Since we shut down PE1 and PE2 the only available route is:
  Client 1 ↔ CE1 ↔ PE2 ↔ RR ↔ PE4 ↔ CE2 ↔ Server 1
  And that is exactly what we have got as a result of traceroute command.

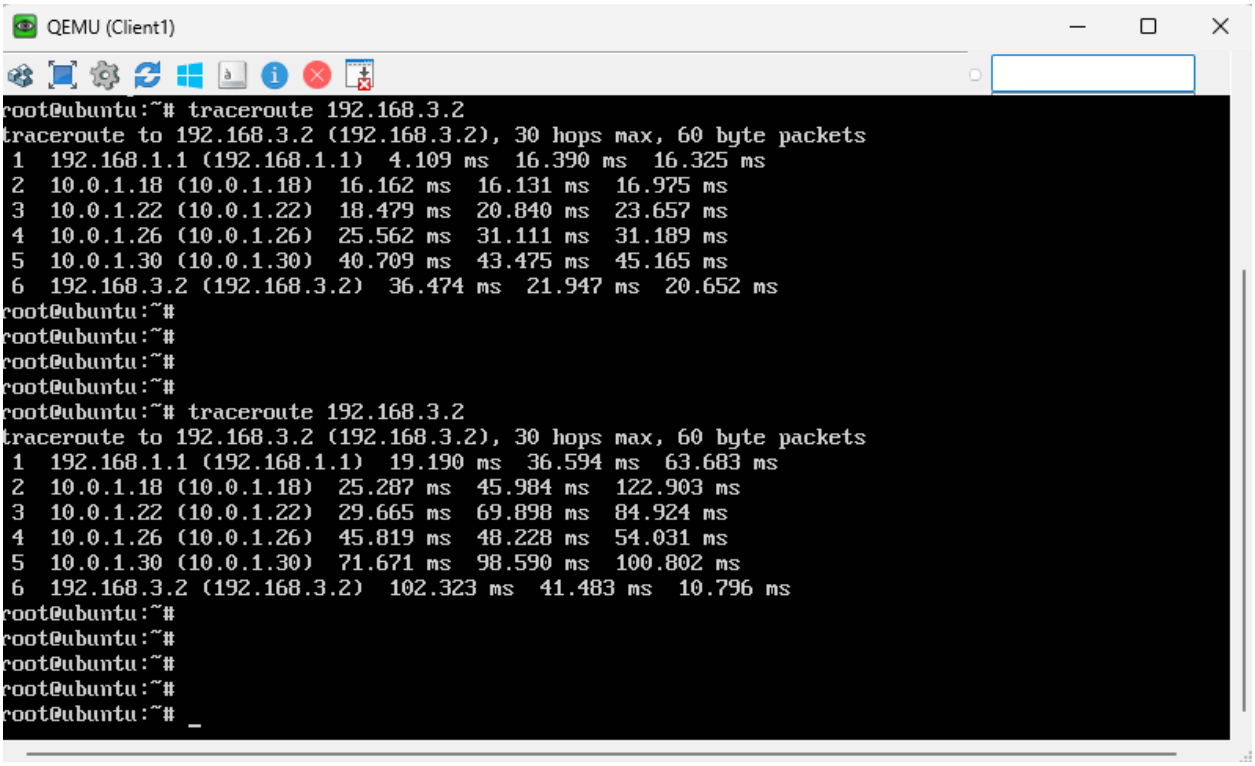


Figure 25: traceroute 3

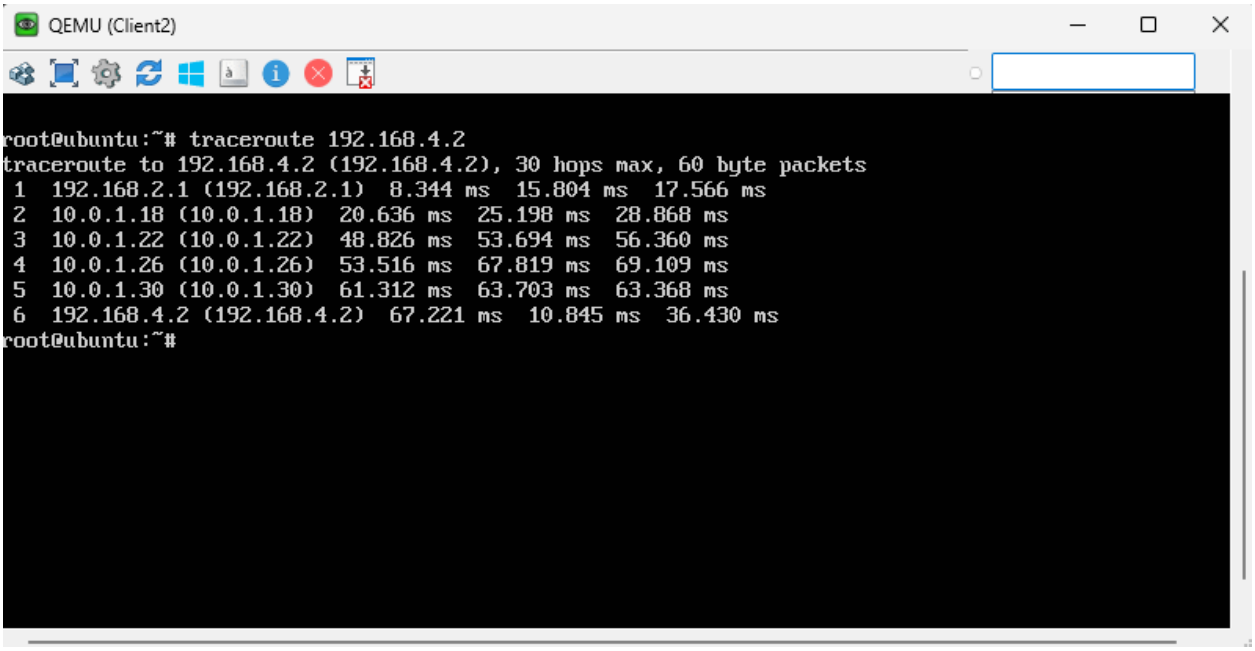


Figure 26: traceroute test 2

IV.10 Summary

The implementation successfully integrates all the major routes and transport technologies used in modern service provider networks:

- ✓ IP routing ensures baseline connectivity.
- ✓ OSPF provides dynamic intra-AS reachability.
- ✓ MPLS introduces label-based transport efficiency.
- ✓ BGP (iBGP/eBGP) handles scalable inter-domain routing.
- ✓ BGP path control fine-tunes traffic behavior across multiple links.

This setup not only simulates real ISP backbone operations but also provides a scalable foundation for future extensions such as L3VPNs, Traffic Engineering, or QoS-based policies.

Chapter V: Perspectives

Although the topology and services deployed were fully functional and aligned with service provider design principles, several enhancements can be developed in future work to align the project more closely with real-world Telco Cloud operations:

1. Deployment of MPLS Layer 3 VPN Services

A natural extension of the current work is the implementation of **Virtual Routing and Forwarding (VRF)** instances and **MP-BGP VPNv4** exchanges. This would:

- ✓ Enable traffic separation between multiple customers.
- ✓ Demonstrate multi-tenant service delivery.
- ✓ Introducing RD/RT management and PE-CE routing models.

2. Integration of Automation Tools

Modern operators rely heavily on automation. Future enhancements may include:

- ✓ Using **Python** for configuration deployment.
- ✓ Utilizing **Ansible** for network orchestration.
- ✓ Creating automatic verification scripts for BGP, MPLS, and path control.

3. Expansion Toward Telco Cloud and NFV Concepts

Extending the project by integrating cloud-native elements such as virtual network functions (vBNG, vFW, vRouter) would provide a deeper understanding of the Telco Cloud environment explored during the internship.

Potential directions include:

- → Exploring container-based networking (Kubernetes CNI/Calico)
- → Deploying VNFs on OpenStack or VMware
- → Investigating service chaining and SDN controllers

Conclusion

This internship project was highly enriching both technically and professionally. It provided an opportunity to work with technologies at the heart of modern service provider infrastructures, deepen understanding of Telco Cloud concepts, and experience real operator engineering workflows.

The skills developed ranging from routing protocol configuration to path engineering, MPLS deployment, and systematic troubleshooting constitute a strong foundation for future roles in network engineering, cloud networking, and telecom architecture.

The perspectives outlined above offer several directions for expanding the project into a more comprehensive service provider lab or integrating it with automation and cloud-oriented tools, which are increasingly essential in today's industry.

Overall, this experience strongly contributed to my professional growth and confirmed my interest in pursuing a career in network and telecommunications engineering.

This internship was an important step in my academic and professional development. It allowed me to move beyond theoretical concepts and understand how large-scale networks are designed, deployed, and operated in a real service provider environment. Working on the IP/MPLS project helped me strengthen my knowledge of routing protocols such as OSPF and BGP, as well as technologies like MPLS that play a key role in modern telecom infrastructures.

Through this project, I built a complete network topology from scratch, implemented core routing protocols, and validated traffic engineering mechanisms such as BGP path control. Reproducing these concepts in EVE-NG and troubleshooting real-world issues taught me how important it is to structure configurations, test methodically, and rely on a solid understanding of how protocols interact with one another. Beyond the technical aspects, it also helped me improve my analytical mindset and my ability to search for solutions independently when facing unexpected behavior.

This internship also gave me the opportunity to discover the world of Telco Cloud, IaaS, and service connectivity, the main domain of my supervisor. This exposure broadened my view of the telecom industry and made me realize how much the sector is evolving toward cloud-based, automated, and virtualized architectures. It confirmed my interest in fields such as networking, cloud engineering, and service provider technologies.

Bibliography

https://www.orange-business.com/en/corporate

https://www.orange-business.com/en/corporate

https://www.greyb.com/blog/6g-companies/

https://www.diskinternals.com/vmfs-recovery/vmware-workstation-and-its-uses/

https://techdocs.broadcom.com/

https://juliopdx.com/2021/08/14/automating-mpls-l3vpns-with-nornir/

https://www.redhat.com/en/topics/cloud-computing/what-is-iaas

https://www.researchgate.net/figure/Traditional-and-SDN-Cloud-Structures_fig1_271850329

https://docs.oracle.com/en/industries/communications/cloud-native-core/2.2.1/policy_user_guide/whats-new-this-guide-pcf-ug.html

https://elysiumpro.in/best-networking-projects/#:~:text=stage%20of%20development.-,Trending%20Topics%20for%20Best%20Networking%20Projects%20in%202025,Real%2DTime%20IoT%20Device%20Monitoring

https://ipwithease.com/sdn-vs-sd-wan/

https://carrier.huawei.com/en/industry-perspective/5g-core-network/next-generation-telco-cloud-based-on-cloud-native

https://www.techtarget.com/searchnetworking/definition/software-defined-networking-SDN

https://networkinterview.com/telco-cloud-architecture/

https://www.redhat.com/en/topics/cloud-native-apps/vnf-and-cnf-whats-the-difference